José Pablo Rodríguez Pinal



Mechatronics Engineering Student · Industrial Automation & Control

Seeking internship with thesis in Germany · Available February – July 2027

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Portfolio · LinkedIn · GitHub

PROFILE

Mechatronics Engineering student at Tec de Monterrey, pursuing a double degree in Smart Automation at Hochschule Harz. Hands-on experience in industrial automation (Rockwell and Siemens PLCs, safety, HMI, FANUC robots, machine vision) and embedded control (STM32), plus six months in automotive R&D at Nemak automating engineering workflows. Looking for an internship with thesis in automation and control.

EDUCATION

Hochschule Harz – Double Degree in Smart Automation Sep 2026 – Jul 2027
Germany · DAAD scholarship · Thesis supervised by Hochschule Harz

Tecnológico de Monterrey – Campus Monterrey Expected Dec 2027
B.S. in Mechatronics Engineering

PROFESSIONAL EXPERIENCE

R&D Engineering Intern – Nemak Mar 2026 – Aug 2026
Research & Development department · Automotive

- **Project proposal automation (Power Apps, Power Automate, SharePoint):** built an end-to-end workflow covering data capture, approvals, business calculations and automatic generation of standard PowerPoint decks and one-pagers; **cut registration time per proposal by 83% (2 h → 20 min).**
- **R&D Knowledge Center (Azure Logic Apps):** automated workflows for the central repository of engineering reports and documentation, reducing repeated searches and duplicated work.
- **Analytics (Power BI):** developed dashboards that turn departmental data into insights for engineering decisions.

Recommendation letter from Dr. José Talamantes Silva, R&D Director (available in portfolio).

SELECTED PROJECTS

Mechatronic Automation System for a Scaled Tractor · *Partner: John Deere* Dec 2026

- CAD design in SolidWorks and electromechanical integration with STM32, IMU, encoders and ultrasonic sensor; trajectory and speed control using fuzzy logic.
- MATLAB interface for parametrized trajectories with real-time telemetry (speed, distance, orientation, obstacle detection).

Automated Manufacturing Cell · *Partner: Rockwell Automation* Jun 2026

- Coordinated Compact GuardLogix PLCs, a GuardLogix safety PLC and EtherNet/IP into one safe, validated sequence for transporting, assembling and inspecting a part.
- Integrated conveyors, pneumatics, FANUC robots and Cognex vision in a multi-team production cell.

Industrial Vision System for Quality Inspection · *Partner: Kemet* May 2026

- Automated detection of printing defects in a manufacturing process using ESP32-CAM.
- Automatic process stop on error detection, reducing material waste and inspection time.

Assistive Robotic Manipulator with Computer Vision · *Partner: INAOE* Jan 2026

- Dynamic modeling (double pendulum approach) and PD control with gravity compensation for safe motion.
- Computer vision (YOLO, Python) for object detection and automatic trajectory execution in RoboDK.

Industrial Automation with Siemens SIMATIC PLC and HMI Oct 2025

- PLC programming (SIMATIC S7, TIA Portal) for a physical sorting system with magnetic sensors.
- HMI for remote crane-type control with real-time monitoring and positioning on a functional prototype.

TECHNICAL SKILLS

PLC & Automation: Siemens SIMATIC S7 (TIA Portal), Allen-Bradley / Rockwell (Studio 5000, Ladder, GuardLogix safety), EtherNet/IP, HMI
Programming: Python, MATLAB, C, C++
Robotics & Vision: FANUC, RoboDK, YOLO, Cognex, ESP32-CAM
Embedded Systems: STM32, ESP32, Arduino
Software & CAD: Simulink, SolidWorks, Emulate3D, FluidSIM, Autodesk Fusion 360, ANSYS Mechanical
Low-code & Data: Power Apps, Power Automate, SharePoint, Azure Logic Apps, Power BI

LANGUAGES, SCHOLARSHIPS & DISTINCTIONS

Spanish: native · **English:** advanced, TOEFL ITP 590 (2025) · **German:** B2, Goethe-Zertifikat (2026)
DAAD Scholarship – study stay at Hochschule Harz, Germany
Bronze Medal – ISSSP International Nano-Satellite Competition 2024, Hong Kong